



Department of Computer Science and Engineering
National Institute of Technology, Rourkela
Mid Semester Examination (Autumn 2025 - 26)
MTech 1st Semester
Sub: Foundations of Computer Security (CS - 6301)
Time: 2 Hrs; FM: 30

Answer **ALL** questions.

1. (a) Define security goals and different security mechanisms. [2] [CO1, BL1]
(b) What is the difference between $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$? How to find the multiplicative inverse of an integer in $\mathbb{Z}_n$? [1 + 2 = 3] [CO1, BL3]
2. (a) Find the gcd of 24 and 320 using the Euclidean algorithm. [2.5] [CO2, BL2]
(b) Using the properties of the mod operator, prove that the remainder of any integer when divided by 10 is the rightmost digit. [2.5] [CO2, BL3]
3. (a) Find the multiplicative inverse of 24 in $\mathbb{Z}_{180}$. [2.5] [CO1, BL2]
(b) Find the solution of a linear Diophantine equation $14x + 21y = 77$. [2.5] [CO3, BL2]
4. (a) Find all solutions of the linear equation $3x + 5 \equiv 4 \pmod{5}$ [2] [CO3, BL2]
(b) Find the solutions to the linear equation

$$7x + 3y \equiv 3 \pmod{7}$$

$$4x + 2y \equiv 5 \pmod{7}$$

[3] [CO2, BL3]

5. (a) Encrypt the message "**We live in an insecure world**" using the Hill cipher and the key **K** as given below.

$$K = \begin{bmatrix} 03 & 02 \\ 05 & 07 \end{bmatrix}$$

[5] [CO2, BL3]

- (b) Decipher the following message using a brute-force attack. Assume that you know it is an affine cipher and that the plain text "**ab**" is enciphered to "**GL**". [5] [CO2, BL3]

XPALASXYFGFUKPXUSOGEUTKCDGFXANMGNVS